NAT & PAT

Course Code: CSC 3116 Course Title: Computer Networks

Dept. of Computer Science
Faculty of Science and Technology

Lecturer No:	11	**Week No:**		**Semester:**	Summer 22-23
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Lecture Outline

1. Private Network
2. NAT
3. NAT Operation
4. Pooling of IP Address
5. PAT
6. Load Balancing of Servers
7. Advantage/Disadvantage

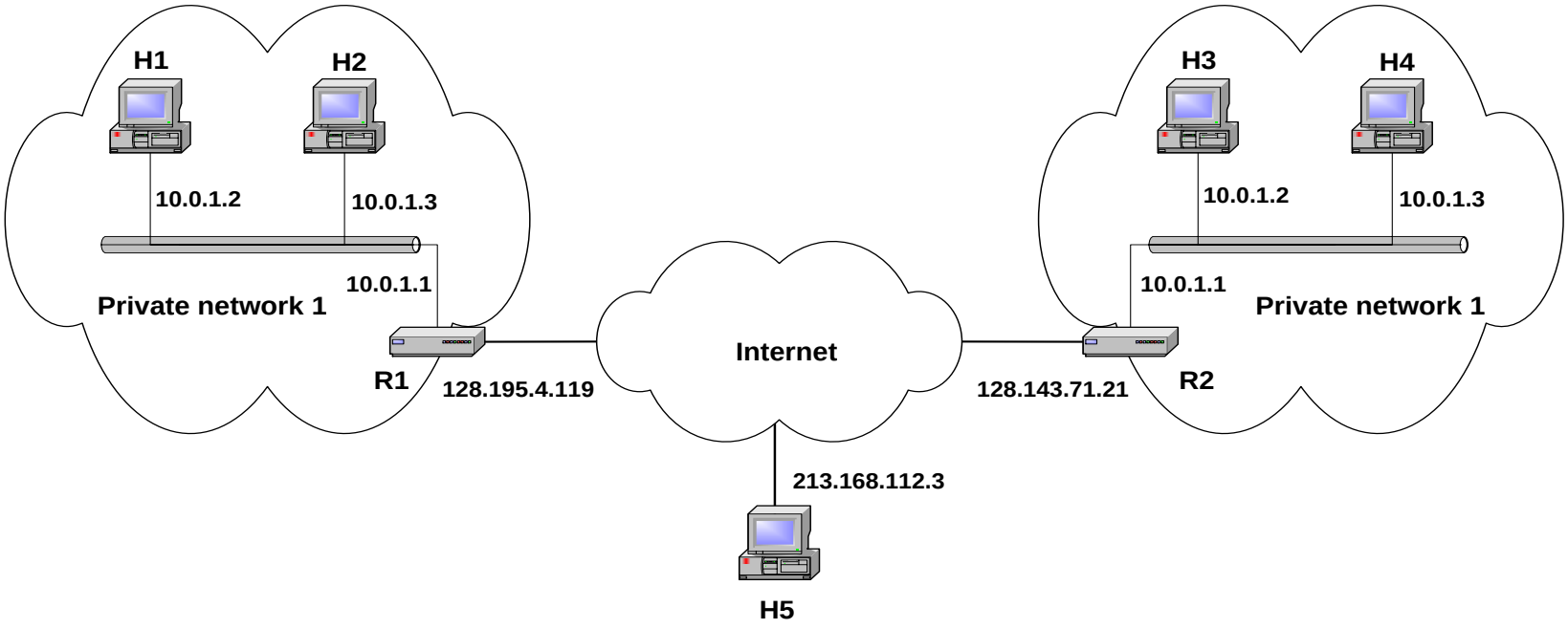
Private Network

Private IP network is an IP network that is not directly connected to the Internet. IP addresses in a private network can be assigned arbitrarily. Not registered and not guaranteed to be globally unique

Generally, private networks use addresses from the following experimental address ranges (*non-routable addresses*):

10.0.0.0 – 10.255.255.255
172.16.0.0 – 172.31.255.255
192.168.0.0 – 192.168.255.255

Private Addresses



Network Address Translation (NAT)

- A short-term solution to the problem of the depletion of IP addresses
- Long term solution is IP v6
- CIDR (Classless Inter Domain Routing) is a possible short-term solution
- NAT is another

NAT is a way to conserve IP addresses
Can be used to hide a number of hosts behind a single IP address
Uses private addresses:

10.0.0.0-10.255.255.255,
172.16.0.0-172.32.255.255
192.168.0.0-192.168.255.255

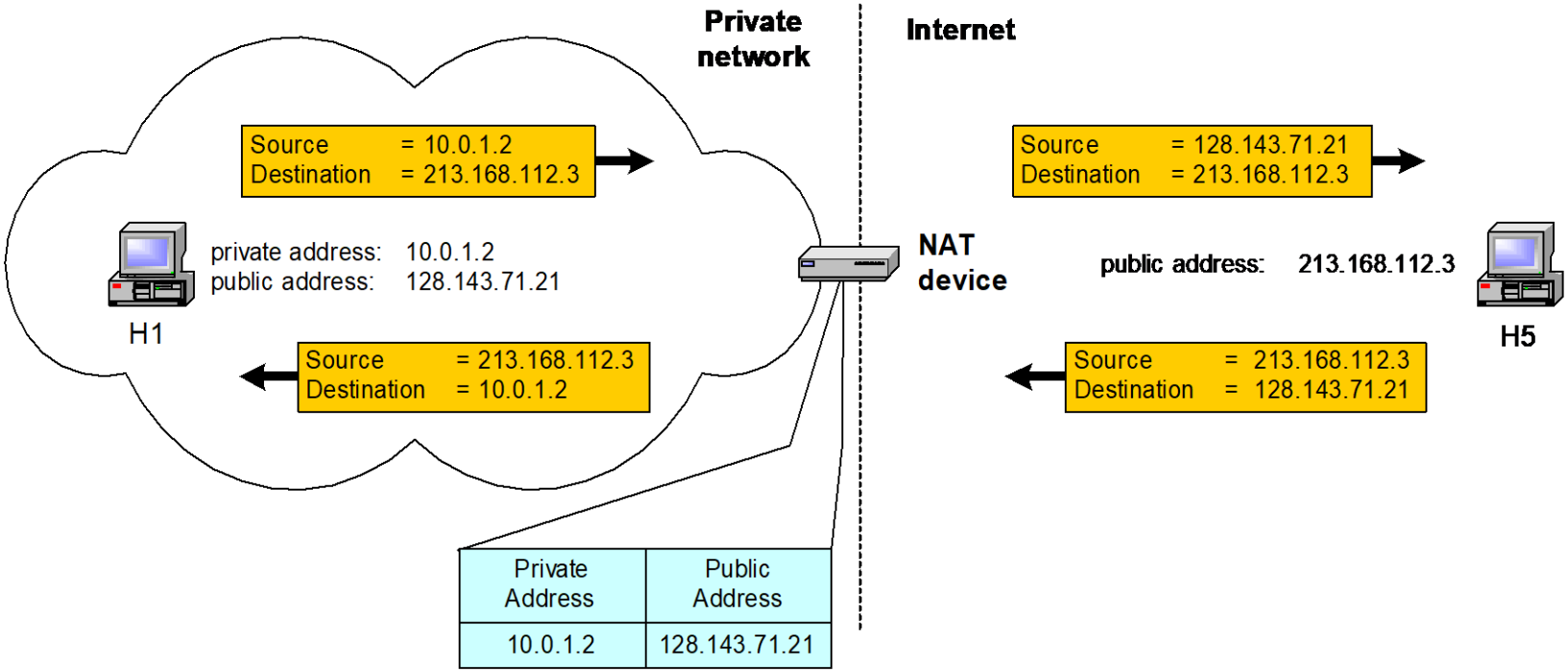
NAT (cont.)

NAT is a router function where IP addresses (and possibly port numbers) of IP datagrams are replaced at the boundary of a private network

NAT is a method that enables hosts on private networks to communicate with hosts on the Internet

NAT is run on routers that connect private networks to the public Internet, to replace the IP address-port pair of an IP packet with another IP address-port pair.

Basic Operation of NAT



NAT device has address translation table

One to one address translation

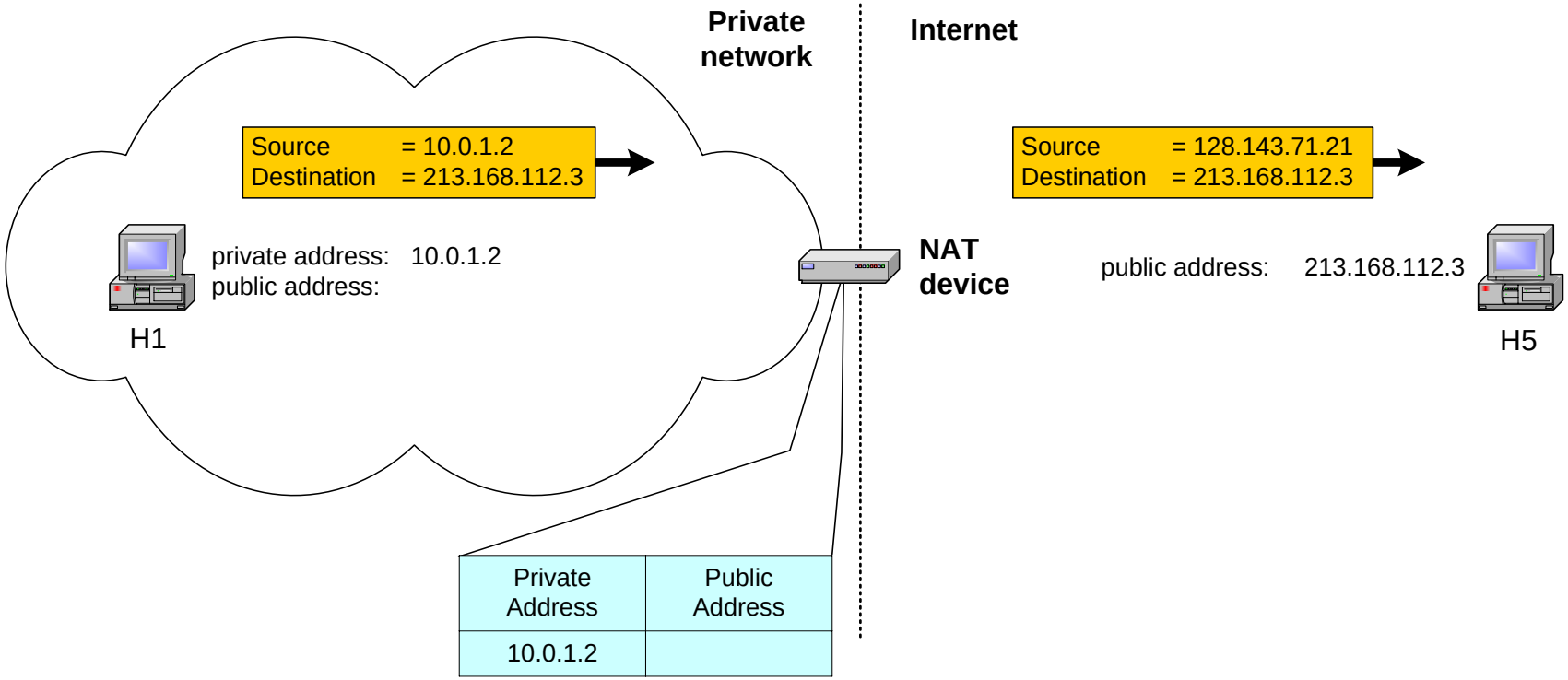
Pooling of IP Addresses

Scenario: Corporate network has many hosts but only a small number of public IP addresses

NAT solution:

- Corporate network is managed with a private address space
- NAT device, located at the boundary between the corporate network and the public Internet, manages a pool of public IP addresses
- When a host from the corporate network sends an IP datagram to a host in the public Internet, the NAT device picks a public IP address from the address pool, and binds this address to the private address of the host

Pooling of IP Addresses (cont…)



Pool of addresses: 128.143.71.0-128.143.71.30

Supporting Migration between Network Service Providers

Scenario: In CIDR, the IP addresses in a corporate network are obtained from the service provider. Changing the service provider requires changing all IP addresses in the network.

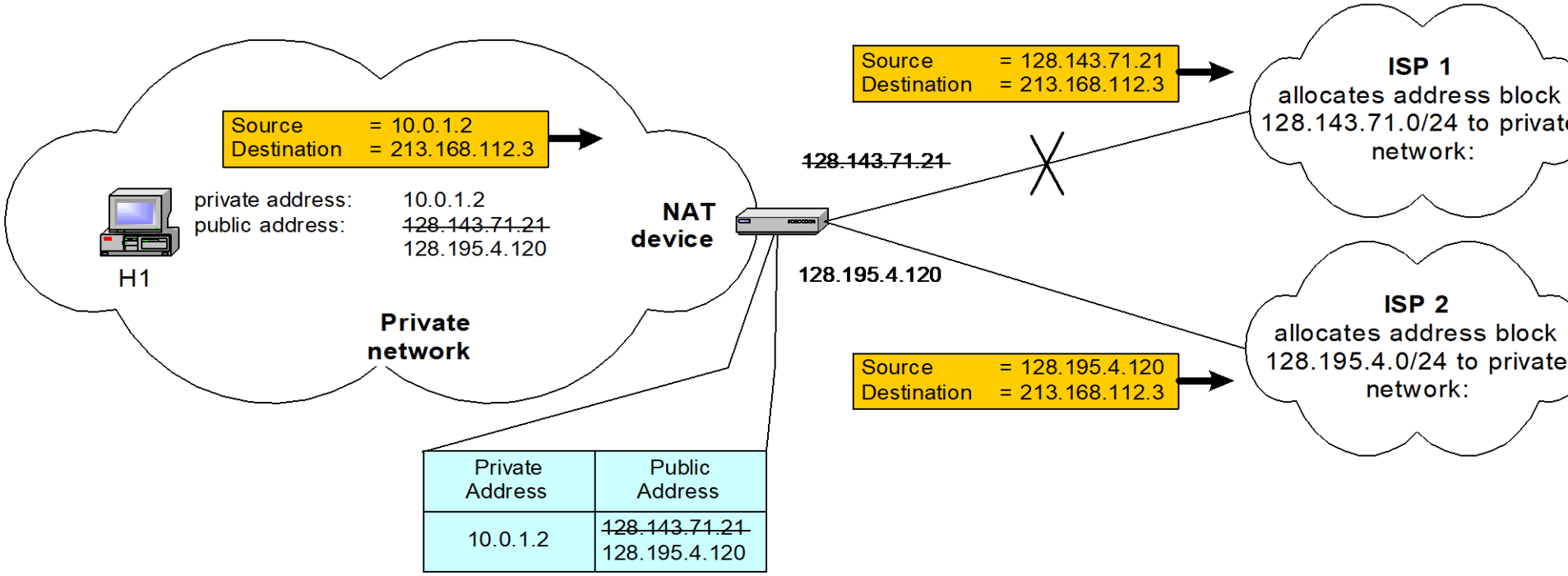
NAT solution:

Assign private addresses to the hosts of the corporate network

NAT device has static address translation entries which bind the private address of a host to the public address.

Migration to a new network service provider merely requires an update of the NAT device. The migration is not noticeable to the hosts on the network.

Supporting Migration between network service Providers (cont…)

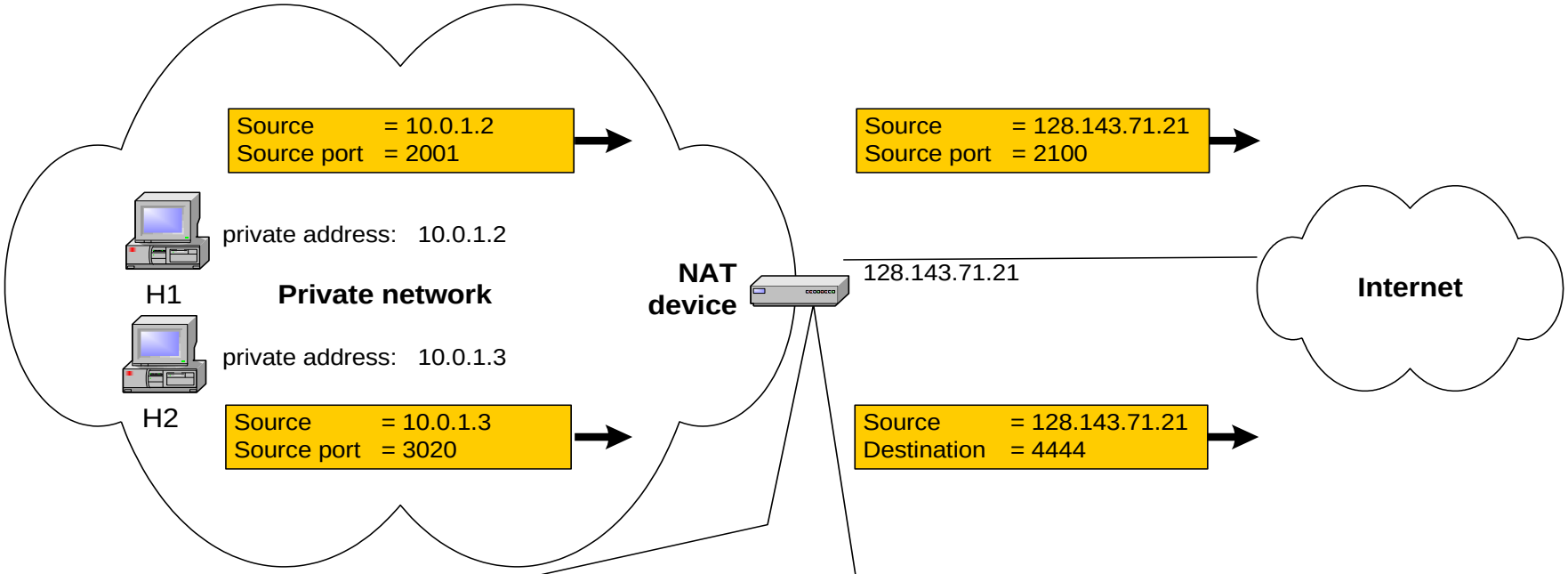


IP Masquerading

Also called: Network address and port translation (NAPT), port address translation (PAT).
Scenario: Single public IP address is mapped to multiple hosts in a private network.
NAT solution:

Assign private addresses to the hosts of the corporate network
NAT device modifies the port numbers for outgoing traffic

IP Masquerading (cont…)



Private Address	Public Address
10.0.1.2/2001	128.143.71.21/2100
10.0.1.3/3020	128.143.71.21/4444

Load Balancing of Servers

Scenario: Balance the load on a set of identical servers, which are accessible from a single IP address

NAT solution:

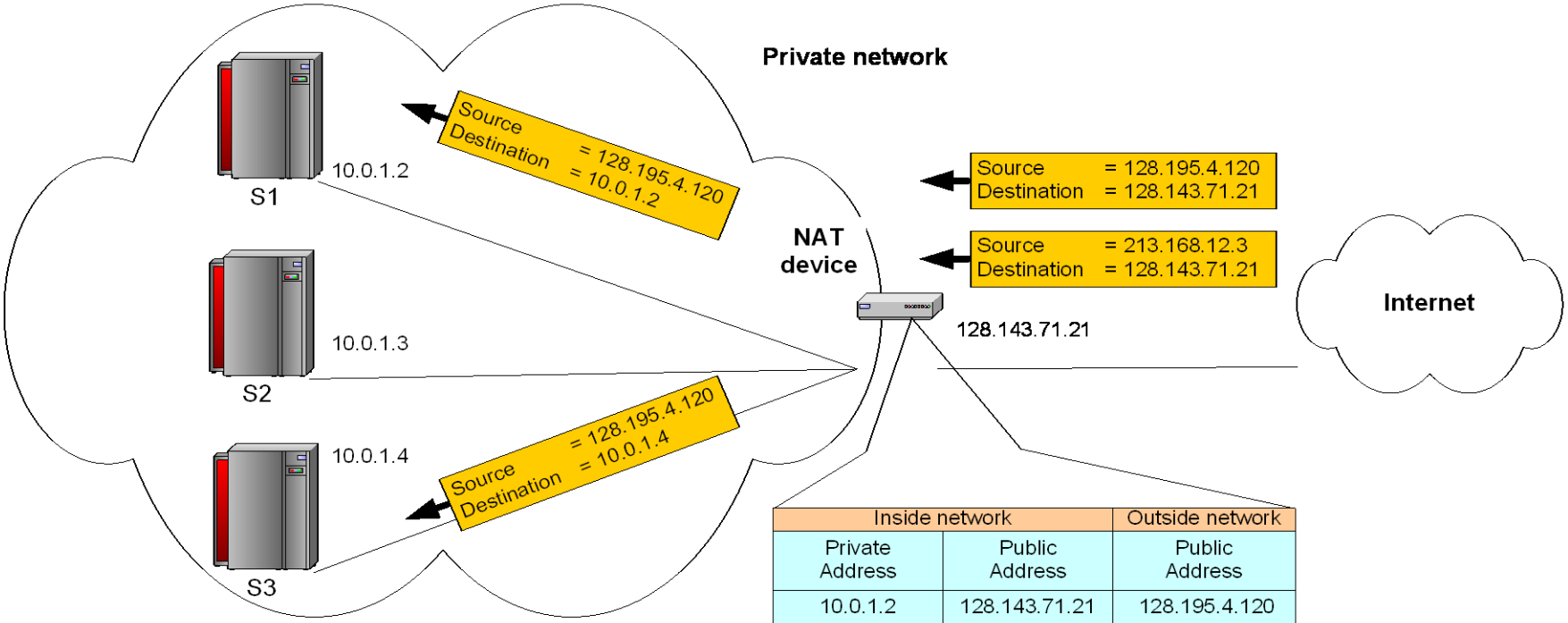
Here, the servers are assigned private addresses

NAT device acts as a proxy for requests to the server from the public network

The NAT device changes the destination IP address of arriving packets to one of the private addresses for a server

A sensible strategy for balancing the load of the servers is to assign the addresses of the servers in a round-robin fashion.

Load Balancing of Servers (cont…)



Inside network		Outside network
Private Address	Public Address	Public Address
10.0.1.2	128.143.71.21	128.195.4.120
10.0.1.4	128.143.71.21	213.168.12.3

Concerns about NAT

Performance:

Modifying the IP header by changing the IP address requires that NAT boxes recalculate the IP header checksum

Modifying port number requires that NAT boxes recalculate TCP checksum

Fragmentation

Care must be taken that a datagram that is fragmented before it reaches the NAT device, is not assigned a different IP address or different port numbers for each of the fragments.

Advantage

Advantages of NAT

• The main advantage of NAT (Network Address Translation) is that it can prevent the depletion of IPv4 addresses.

• NAT (Network Address Translation) can provide an additional layer of security by making the original source and destination addresses hidden.

• NAT (Network Address Translation) provides increased flexibility when connecting to the public Internet.

• NAT (Network Address Translation) allows to use your own private IPv4 addressing system and prevent the internal address changes if you change the service provider.

Disadvantage

Disadvantages of NAT

• NAT (Network Address Translation) is a processor and memory resource consuming technology, since NAT (Network Address Translation) need to translate IPv4 addresses for all incoming and outgoing IPv4 datagrams and to keep the translation details in memory.

• NAT (Network Address Translation) may cause delay in IPv4 communication.

• NAT (Network Address Translation) cause loss of end-device to end-device IP traceability

References

1. **Data Communications and Networking**, *B. A. Forouzan*, McGraw-Hill, Inc., Fourth Edition, 2007, USA.
2. https://www.geeksforgeeks.org/basics-computer-networking/
3. https://www.tutorialspoint.com/computer_fundamentals/computer_networking.htm

Books

1. **Data Communications and Networking**, *B. A. Forouzan*, McGraw-Hill, Inc., Fourth Edition, 2007, USA.
2. **Computer Networking: A Top-Down Approach**, *J. F., Kurose, K. W. Ross*, Pearson Education, Inc., Sixth Edition, USA.
3. **Official Cert Guide CCNA 200-301 , vol. 1**, *W. Odom*, Cisco Press, First Edition, 2019, USA.
4. **CCNA Routing and Switching**, *T. Lammle*, John Wily & Sons, Second Edition, 2016, USA.
5. **TCP/IP Protocol Suite**, *B. A. Forouzan*, McGraw-Hill, Inc., Fourth Edition, 2009, USA.
6. **Data and Computer Communication**, *W. Stallings*, Pearson Education, Inc., 10th Education, 2013, USA.